

Problem Elaboration:

Public restrooms are not designed with young children's physical abilities in mind. Standard stall locks and latches are typically positioned high on the door¹, a range intended for adults and not children's reach. For children aged 3-4, recommended operable reach heights are no more than 36 inches, which means many stall locks are outside a young child's comfortable reach².

In our case, Mr. Kiser and his four-year-old child are going on vacation and will not have access to a private restroom. Thus, they need a public restroom lock as the child cannot reach the stall lock due to its height and/or complexity. Because the child cannot operate the lock, they are at risk for accidents, unsanitary conditions, and the fear of being watched while using the restroom. This problem affects the child's privacy, independence, and comfort whilst increasing parental stress and travel inconvenience³.

Although our lock will center on Mr. Kiser's needs, many caregivers face a similar challenge aiding children in their use of the restroom. A solution would not only help families on vacation, but also teachers of young children, caregivers, and childcare providers. Addressing this issue would also aid inclusive design as a whole, where individual needs are prioritized for the betterment of society.

Problem Identification:

The problem is to create a device that a parent can use outside of the bathroom stall that can be used to lock the door shut for a 15-minute time period without the need for child operability. The door must be perpendicular to the wall. The lock takes no more than 10 seconds to remove and put on. The lock needs to be lightweight, follow TSA requirements, and fit into the parent's carry-on luggage or pockets (around 9 x 6.4)⁴.

Stakeholder Concerns and Connections:

Primary Stakeholders:

Mr. Kiser is our primary stakeholder as he is going on a vacation with his four-year-old, who cannot use the restroom on their own.

Parents/Guardians: They are responsible for their child's safety and comfort.

Their concern is to ensure that the child is secure and that the bathroom stop is as smooth as possible with minimal effort (ex. The carrying weight).

Caregivers/Childcare providers: They need reliable solutions that do not rely on holding doors manually, as they need to care for multiple children.

Young children: They cannot reach or operate existing stall locks and need assistance.

Secondary Stakeholders:

Facility Managers: They benefit from safer restroom usage and reduced property damage from ineffective lock solutions⁵.

Other restroom users: They benefit from a quick solution to the problem as they need to use the stalls as well⁶.

Problem Justification:

Current restroom locks were designed for general adult use and often do not account for the ergonomic limitations of young children⁷. Guidelines note that children's forward reach for operable parts should remain at or below about 36 inches tall (for ages 3-4), but many stall locks exceed this range, making them difficult for children to interact with independently. This problem occurs approximately 4-7 times per day as the child needs to use the restroom⁸. The inability to secure a stall lock independently may lead to interruptions during use, compromise privacy, and upset children, especially when traveling and public restrooms are the only choice. Addressing this problem directly with a pocket portable, parent-operated lock that works on a wide range of standard stall doors would improve safety, usability, and comfort for children without requiring structural changes to bathrooms on a wide scale. The lock solution must not include the child, as that may put undue stress on them, and create the same issues they have in

Citations

1. (34-38 inches off the floor) Ironwood Mfg, 9-19-2025, "ADA Toilet Partition Door Requirements", <https://ironwood-mfg.com/blog/ada-design-toilet-partition-doors/>
2. (Comfortable Reach) ADA, 7-23-2004, "ADA-ABA Accessibility Guidelines", No Publication, https://archive.ada.gov/archive/NPRM2008/ada_standards/proposedadastds.htm
3. (Open Restroom Distress) NSPT, 2-12-2024, "The Dirty Little Secret on Poop – Tips For Children Who Won't Use the Potty in Public or at School", No Publication, <https://www.nspt4kids.com/parenting/the-dirty-little-secret-on-poop---tips-for-children-who-wont-use-the-potty-in-public-or-at-school>

4. (Pocket Size) Dmassey, 2-15-2021, "The Power of Pockets", Kendal Corporation,
<https://www.kendal.org/2021/02/15/the-power-of-pockets-their-history-meaning-through-the-years/>
5. (Facility Managers and their jobs rely on public restroom well-being) Strong Towns, 07-07-2016, "Let's get serious about public bathrooms", Strong Towns Archive,
<https://archive.strongtowns.org/journal/2016/7/1/public-bathrooms>
6. (Quick visits are optimal) Christine B. Whittemore, 7-24-2017, "Bathrooms Affect Retail Experience Say Studies", Simple Marketing,
<https://www.simplemarketingnow.com/blog/flooring-the-consumer/bid/137884/bathrooms-affect-retail-experience-say-studies>
7. (Restrooms were not built for comfortable child use) Emma Durand-Wood, 05-16-2024, "Dignity, Inclusivity and Walkability: How Public Bathrooms Help Cities Thrive", Strong Towns Archive,
<https://archive.strongtowns.org/journal/2024/5/16/dignity-inclusivity-and-walkability-how-public-bathrooms-help-cities-flourish>
8. (Bathroom visits per day) Children's Dayton 06-30-2025, "When to Be Concerned", Children's Dayton,
<https://childrensdayton.org/resources/when-to-be-concerned/frequent-urination-in-kids/>